

Air Quality Analysis Narrative

Air is life and many people find that good air quality is essential for a high standard of living. Air quality can affect a person's health in a variety of ways both physically and mentally. Many things can impact our air quality, industry, location, weather patterns, and even climate change. Thus, the EPA created the AQI (air quality index) to help give guidelines and structure to how air quality is measured. In this project we will look at the AQI's across the United States from the years 2016-2025.

When looking at the EPA data, the daily readings were well over 1 GB so my analysis comes from the annual summary of the data sorted by CBSA's (Core Based Statistical Area). The EPA defines CBSA's as "a geographic area that consists of one or more counties anchored by an urban center of at least 10,000 people, plus adjacent counties that are socioeconomically tied to the center" The questions that I wanted answered were:

How does geographic location affect AQI scores?

What pollutants affect AQI scores the most?

How have the AQI scores changed over time?

I understand that there are some issues with my methodology that might introduce mathematical biases into my results because of the data I chose if I were to do this project again I would limit my scope and use a smaller time range and use the daily data instead of the annual data. My findings as I understand them are that areas along the West coast tend to have higher AQI medians and maxes. Causes could be extreme weather events like Forrest fires. The pollutants that affected the AQI's the most where PM2.5 and Ozone.